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Satellite Communication - TTCM Subsystem

Telemetry, Tracking, Commanding and Monitoring (**TTCM**) subsystem is present in both satellite and earth station. In general, satellite gets data through sensors. So, Telemetry subsystem present in the satellite sends this data to earth station(s). Therefore, TTCM subsystem is very much necessary for any communication satellite in order to operate it successfully.

It is the responsibility of satellite operator in order to control the satellite in its life time, after placing it in the proper orbit. This can be done with the help of **TTCM subsystem**.

We can make this TTCM subsystem into the following **three parts**.

- Telemetry and Monitoring Subsystem
- Tracking Subsystem
- Commanding Subsystem

Telemetry and Monitoring Subsystem

The word **Telemetry** means measurement at a distance. Mainly, the following operations take place in Telemetry.

- Generation of an electrical signal, which is proportional to the quantity to be measured.
- Encoding the electrical signal.
- Transmitting this code to a far distance.

Telemetry subsystem present in the satellite performs mainly two functions –

- receiving data from sensors, and
- transmitting that data to an earth station.

Satellites have quite a few sensors to monitor different parameters such as pressure, temperature, status and etc., of various subsystems. In general, the telemetry data is transmitted as FSK or PSK.

Telemetry subsystem is a remote controlled system. It sends monitoring data from satellite to earth station. Generally, the **telemetry signals** carry the information related altitude, environment and satellite.

Tracking Subsystem

Tracking subsystem is useful to know the position of the satellite and its current orbit. Satellite Control Center (**SCC**) monitors the working and status of space segment subsystems with the help of telemetry downlink. And, it controls those subsystems using command uplink.

We know that the **tracking subsystem** is also present in an earth station. It mainly focusses on range and look angles of satellite. Number of techniques that are using in order to track the satellite. For **example**, change in the orbital position of satellite can be identified by using the data obtained from velocity and acceleration sensors that are present on satellite.

The **tracking subsystem** that is present in an earth station keeps tracking of satellite, when it is released from last stage of Launch vehicle. It performs the functions like, locating of satellite in initial orbit and transfer orbit.

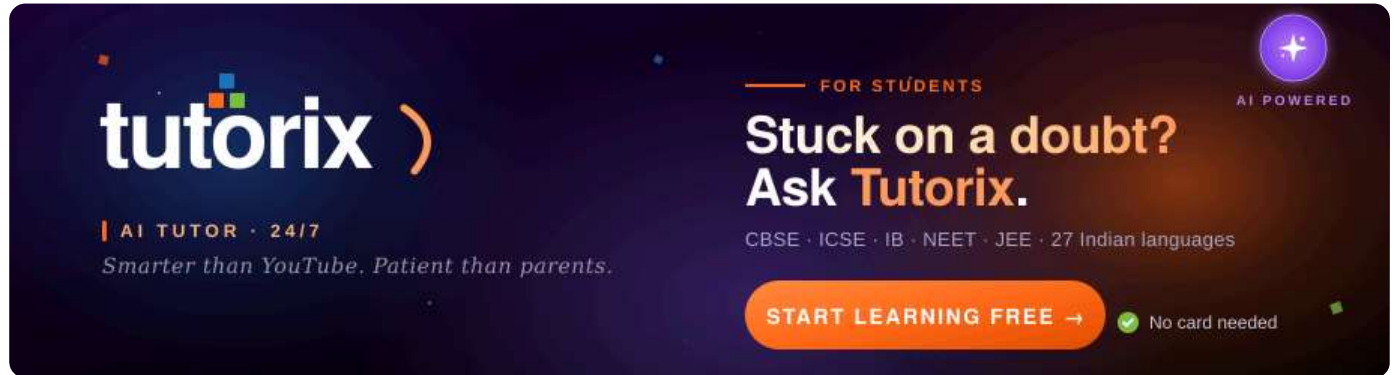
Commanding Subsystem

Commanding subsystem is necessary in order to launch the satellite in an orbit and its working in that orbit. This subsystem adjusts the altitude and orbit of satellite, whenever there is a deviation in those values. It also controls the communication subsystem. This **commanding subsystem** is responsible for turning ON / OFF of other subsystems present in the satellite based on the data getting from telemetry and tracking subsystems.

In general, control codes are converted into command words. These command words are used to send in the form of **TDM frames**. Initially, the validity of command words is checked in the satellite. After this, these command words can be sent back to earth station. Here, these command words are checked once again.

If the earth station also receives the same (correct) command word, then it sends an execute instruction to satellite. So, it executes that command.

Functionality wise, the Telemetry subsystem and commanding subsystem are opposite to each other. Since, the first one transmits the satellites information to earth station and second one receives command signals from earth station.



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